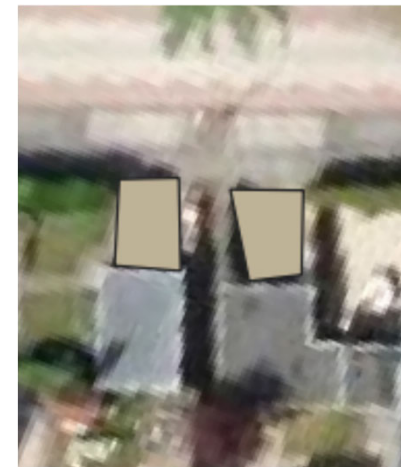


Detectie van privéparkeerplaatsen

Deep Learning project

- Herkennen van (privé)parkeerplaatsen ten behoeve van inrichting leefomgeving
- Sattelietbeelden van enkele wijken als dataset
- Selecteren, trainen, valideren en optimaliseren van verschillende detectiemodellen
- Integratie in Tygronplatform
- Leerdoelen
 - Inhoudelijk: toepassing kennis & vaardigheden Deep Learning
 - Procesmatig: groepsproject voor echte opdrachtgever
 - Toegepast onderzoek



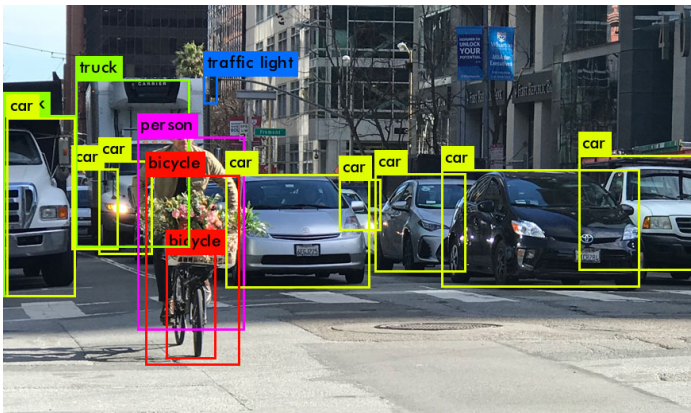
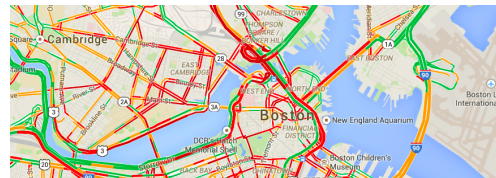
Artificial Intelligence bij HBO-ICT

Onderwijs en projecten

Tijmen Joppe Muller



Artificial Intelligence



Uitstroomprofielen

- Artificial Intelligence Engineer
- Artificial Intelligence Consultant
- Machine Learning Engineer
- Data Scientist
- Data Analyst
- Big Data Consultant
- Software Engineer

HBO-ICT – Artificial Intelligence

Jaar 1	Foundations of ICT	Fundamentals of AI
Jaar 2	Data Science	Deep Learning
Jaar 3	Multidisciplinair project (INNO)	Stage
Jaar 4	Minor	Afstuderen

Fundamentals of AI

Inhoud

Programmeren

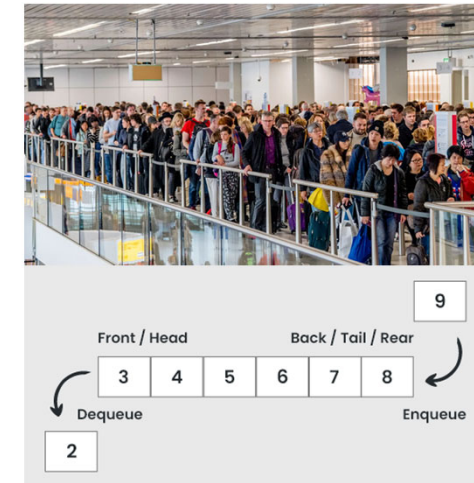
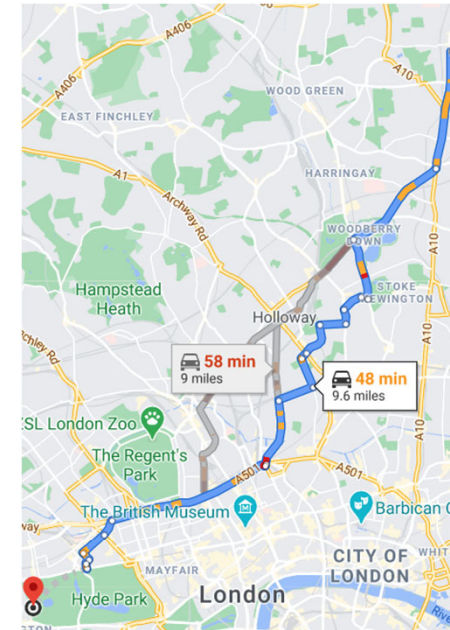
- Procedureel programmeren
- Object-georiënteerd programmeren
- Python & C++
- Algoritmiek
- Datastructuren
- Coding skills

Theorie

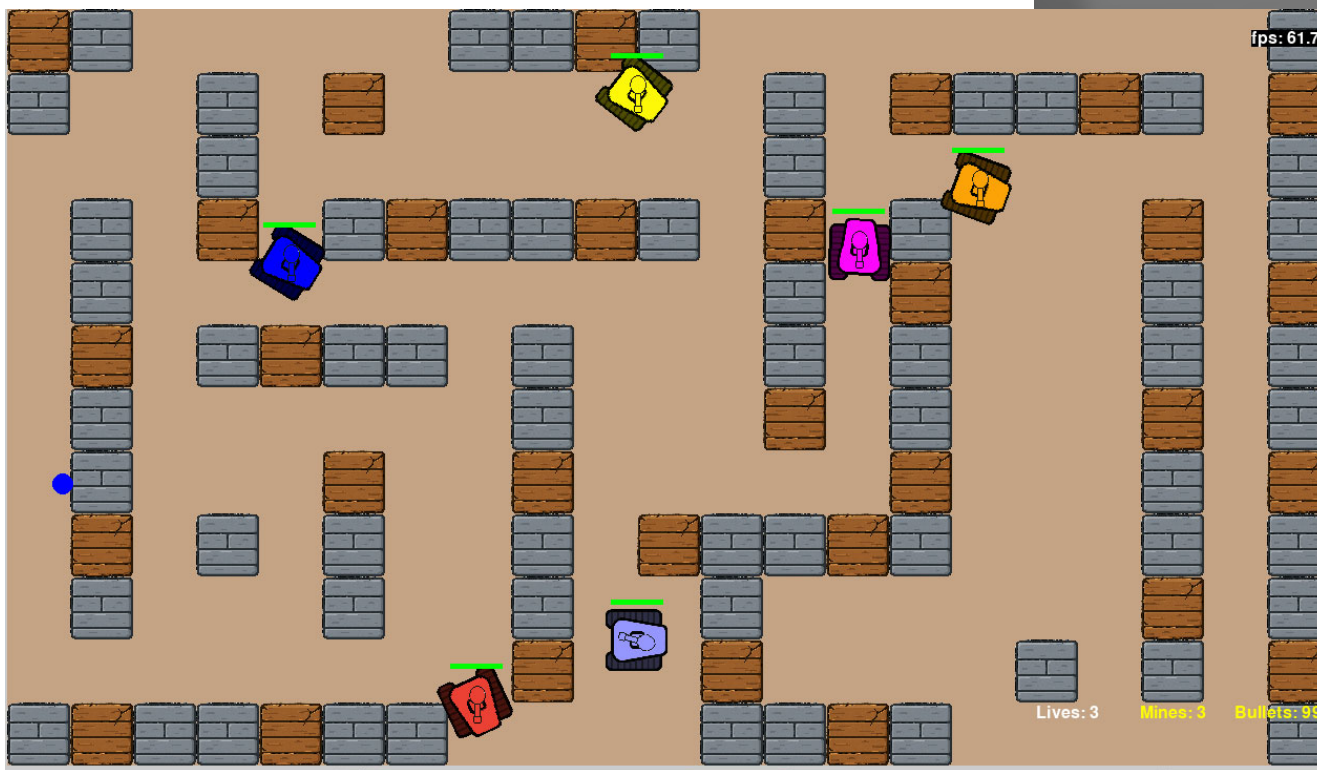
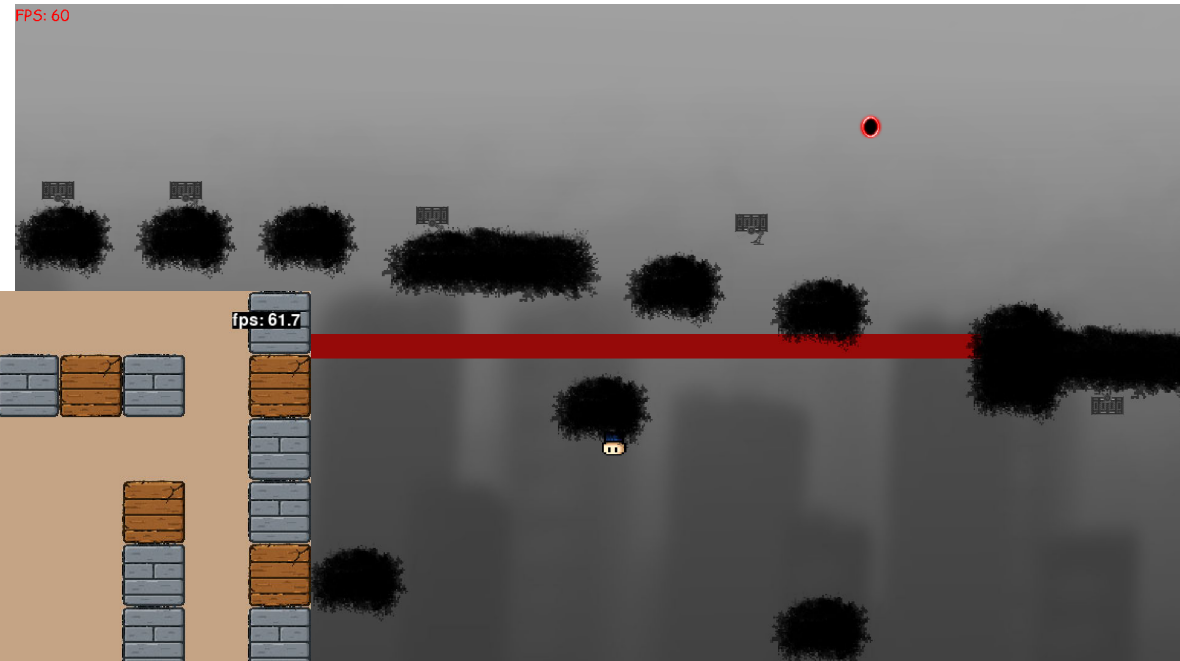
- Logica
- Verzamelingenleer
- Complexiteit van algoritmes
- Hardware-architectuur

Project

- Game development
- Toepassing van algoritmen en heuristieken
- Agile werken



Fundamentals of AI Projecten



Data Science

Inhoud

Programmeren

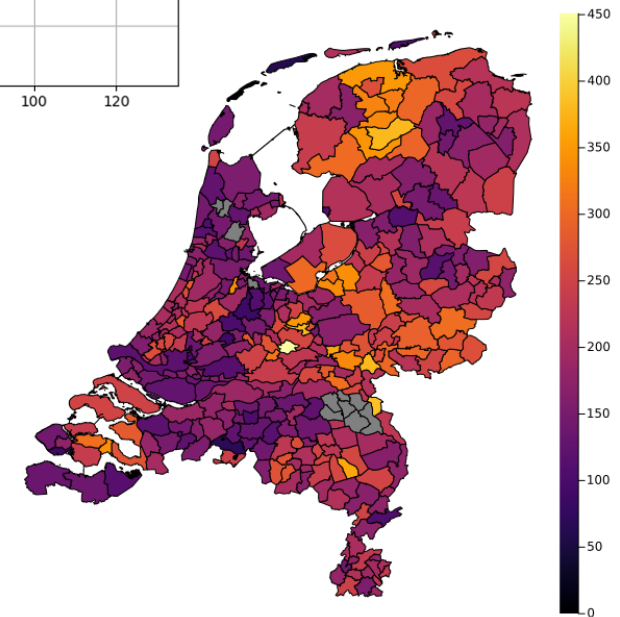
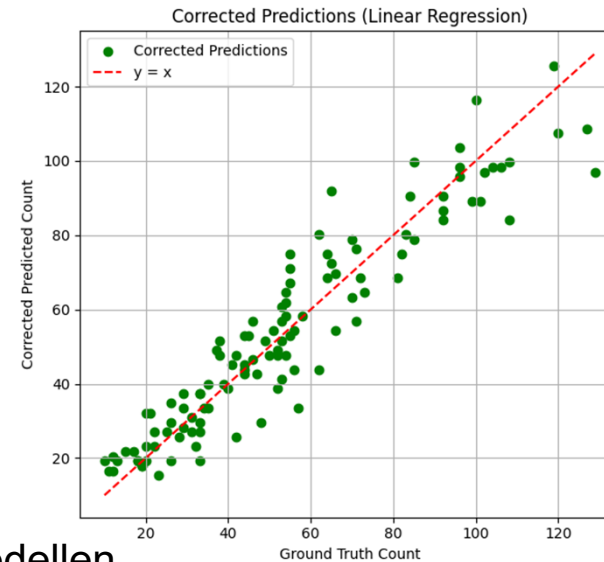
- Datamanipulatie
- Machine learning modellen trainen
- Model deployment
- Numpy, Pandas, Scikit-learn

Theorie

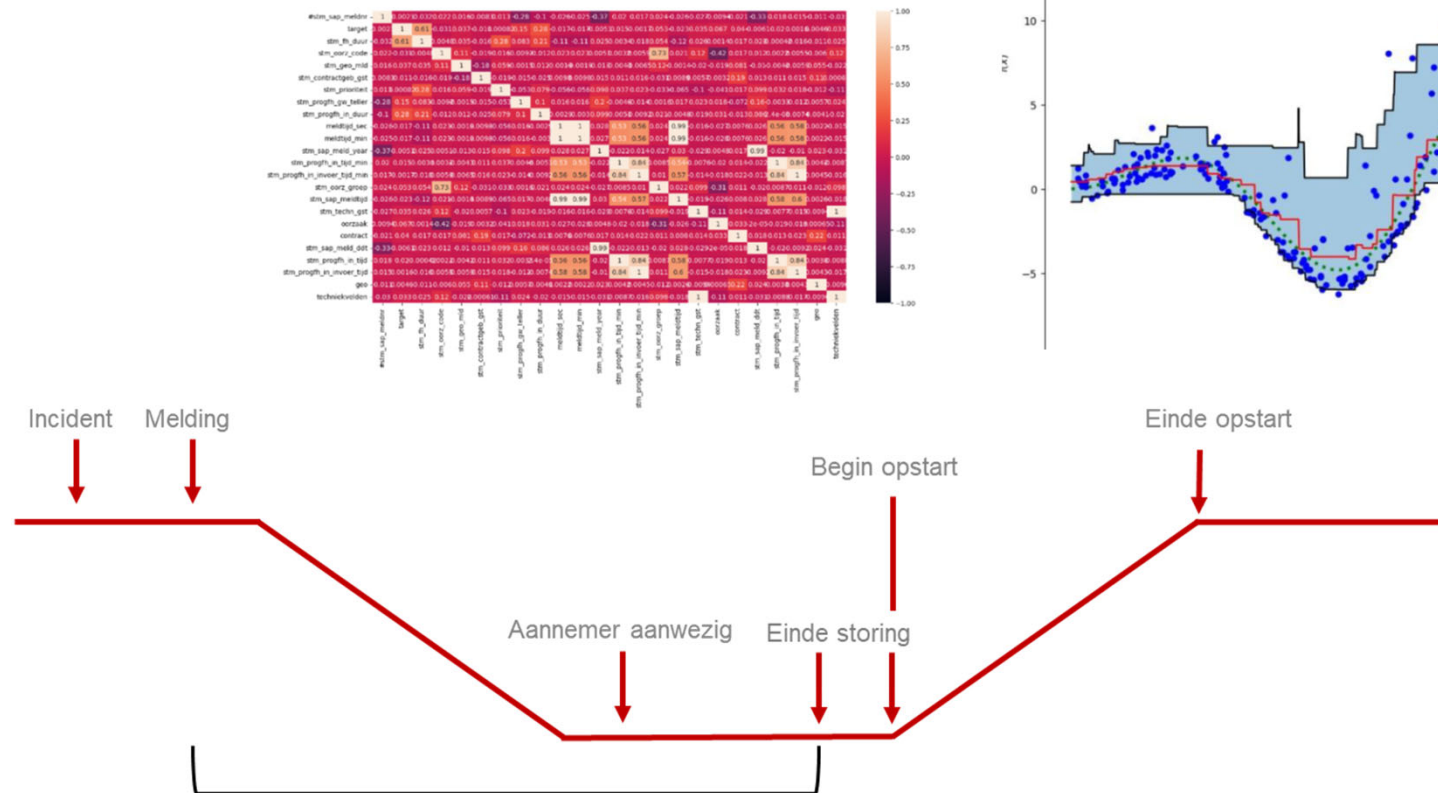
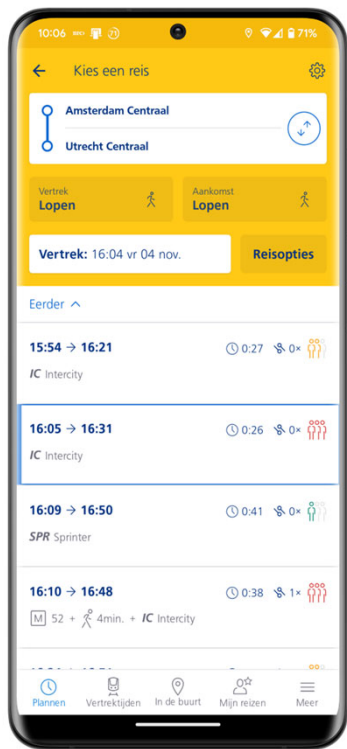
- Statistiek
- Differentiëren
- Machine Learning modellen analyseren

Project

- Data-driven business
- Datavisualisatie en dashboards
- Data en ethiek



Data Science Projecten



Deep Learning

Inhoud

Programmeren

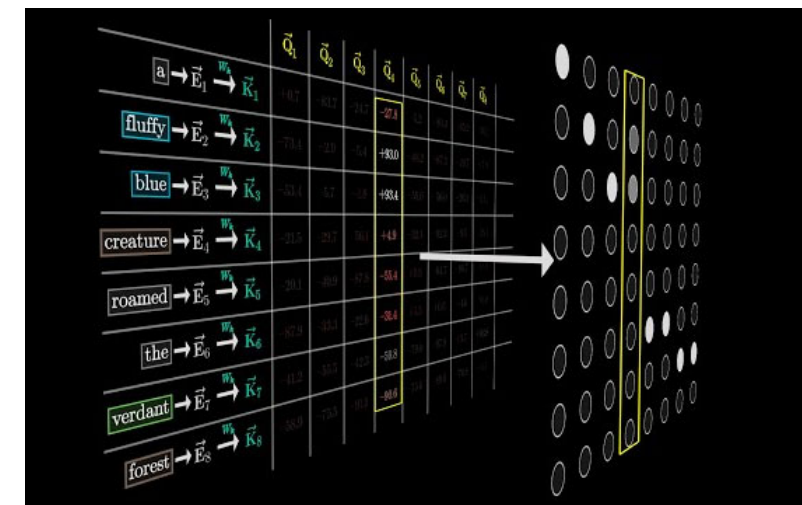
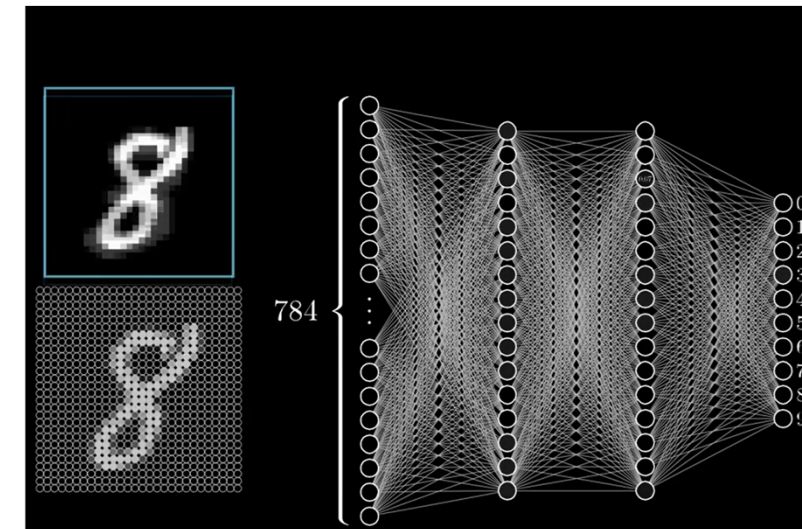
- Neurale netwerken
- Convolutionele netwerken
- Transformermodellen
- Reinforcement learning
- Pytorch & CUDA

Theorie

- Lineair algebra
- Backpropagation
- Convoluties
- Vector embeddings
- Markov modellen

Project

- Data-analyse
- Model trainen & evalueren
- Toegepast onderzoek



Fundamentals of AI

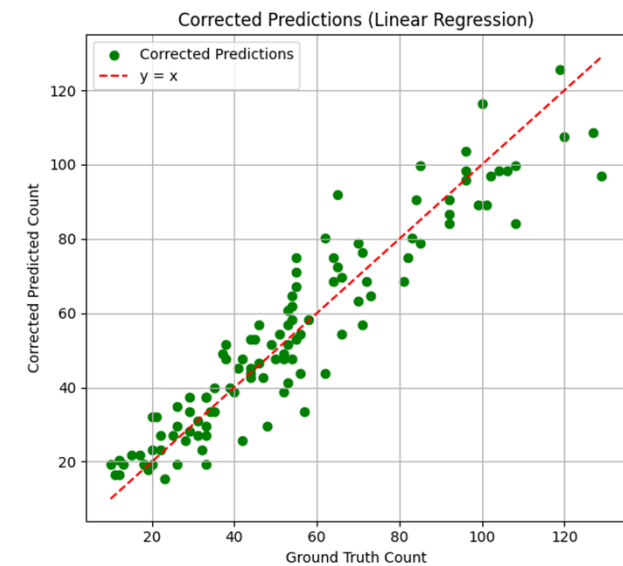
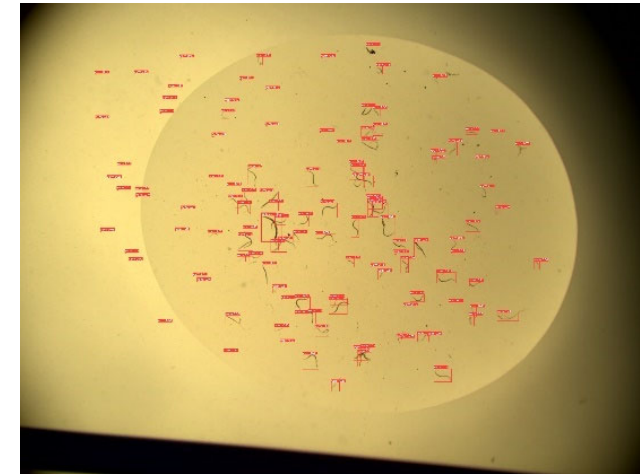
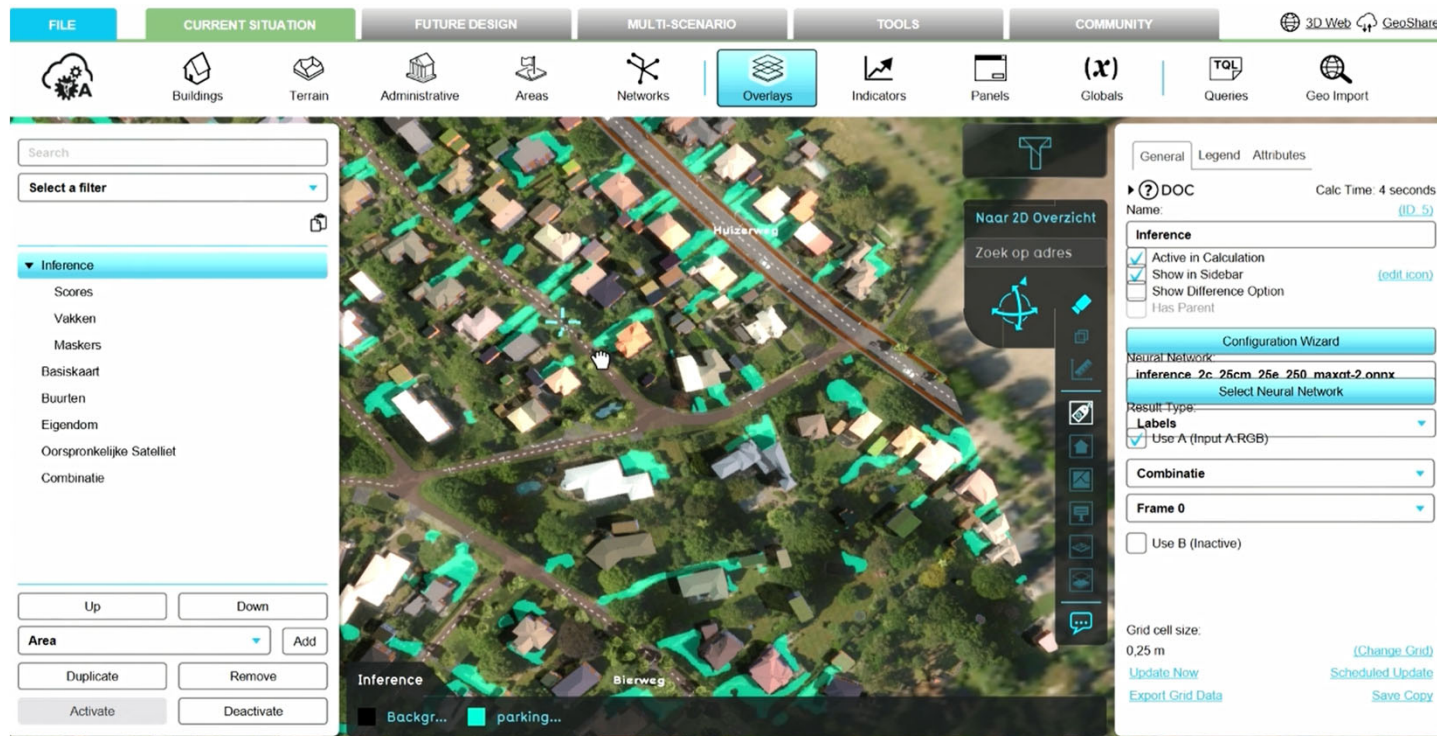
Data Science

Deep Learning

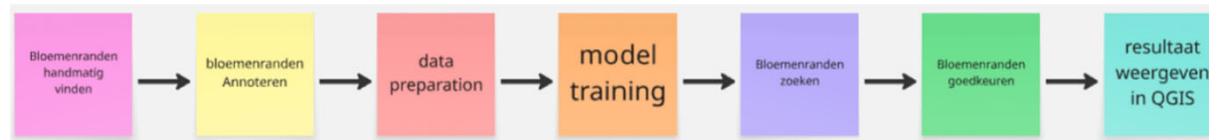
INNO



Deep Learning Projecten



Deep Learning Projecten

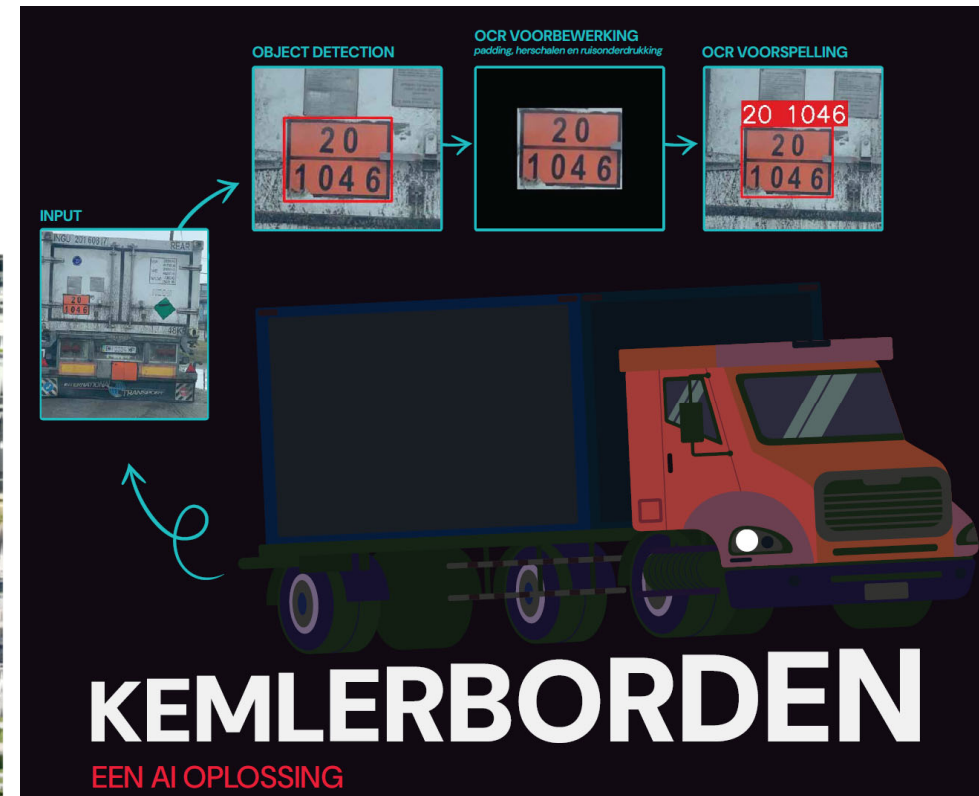


		EFTCO Cleaning Document ® D / 760/D - RV25019757 www.eftco.org DVTI www.dvti.de	
HDK M. Kroker GmbH Im Meer 6-8 · 28816 Stuhr Telefon 0 42 06 14 25 · Telefax 0 42 06-16 15 info@hdk-kroker.de			
2 Kunden Referenznummer / Customer reference number*		3 Serien Nummer / Serial number* 118566	
4 Kunden Daten / Customer Data K000000481 FTC Polska Transport SP.Zo.o. Fabryczna 18 88-210 Dobrze		5 Behälter-Daten / Identification numbers Fahrzeug / Vehicle WPK 5607 P Auflieger, Silos, Container, IBC / Tank, Silo, Container, IBC OKA 7B YA	
6 Art des Produktes / Nature of product*		7 Nächste Befüllung / Next Load*	
8 Letztes Ladegut / Previous load K LTN Nr Name / Name		9 Durchgeführte Arbeiten Carrying Procedures EFTCO Code / Description	
		WKP,P10,C01 WKP,P10,C01 WKP,P10,C01	
K 10 Zusätzliche Arbeiten / Additional Services*			
1 Reinigung 2 Treibstoffe 3 Sammlung der Luft und Spülbehälter 4 Schmelze		113,614,171,181,191,201,211,221,231,241,251,261,271,281,291,301,311,321,331,341,351,361,371,381,391,401,411,421,431,441,451,461,471,481,491,501,511,521,531,541,551,561,571,581,591,601,611,621,631,641,651,661,671,681,691,701,711,721,731,741,751,761,771,781,791,801,811,821,831,841,851,861,871,881,891,901,911,921,931,941,951,961,971,981,991,1001	
11 Bemerkungen / Comments: Vorprodukte & Kundengeräte: Apfelöl, Palm öl, Raps öl, Plomber: 6253 bis 6259 (E90)			
12 Name des Reinigers / Name cleaner*		13 Datum / Time In* 03.03.2025 08:29:38 03.03.2025 11:42:09	
Die Reinigungsstation und die Fahrer bestätigen die oben aufgeführten Leistungen. Der Fahrer hat auch nach EFTCO Definition. The cleaning station and the driver confirm that the above services have been carried out (see EFTCO definition of "clean").		Ende der Reinigung / Time Out	
HDK M. KROKER GmbH Tank-Raumfahrt AG Station Name / Name Tel. 042 06 14 25 Unterschrift / Signature		15 Fahrer / Driver* Name / Name Unterschrift / Signature	

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{ "lpcnr": "OP83VT",  
  "cleaning_station": "OTT Cleaning",  
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Multidisciplinair project Projecten

- Multidisciplinair project (alle afstudeerrichtingen)
- Begeleiding in proces, inhoud en vanuit het werkveld



Vragen?



`tijmen.muller@hu.nl`